$$g_\lambda(Y) = \overbrace{\beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}}^{\mu} + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

What is the distribution of Y?

let $W = g_\lambda(Y)$ such that $W \sim N(\mu, \sigma^2)$

Then $\mathbb{P}(Y \le y) = \mathbb{P}(g_\lambda(Y) \le g_\lambda(y)) = \mathbb{P}(W \le g_\lambda(y))$

$$= \Phi\left(\frac{g_\lambda(y) - \mu}{\sigma}\right)$$

$$\frac{d}{dy}\mathbb{P}(Y \le y) = \varphi\left(\frac{g_\lambda(y) - \mu}{\sigma}\right) \cdot \frac{1}{\sigma} \cdot g_\lambda'(y)$$

STAT2003 claim 28.

$$= (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(g_\lambda(y) - \mu)^2}{2\sigma^2}\right) \cdot y^{\lambda-1}$$

Likelihood

$$\mathcal{L}(\underline{\beta}, \sigma^2, \lambda) = \prod_{j=1}^{n} (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(g_\lambda(y_j) - \underline{x}_j\underline{\beta})^2}{2\sigma^2}\right) y_j^{\lambda-1}$$

log likelihood

$$\ell(\underline{\beta}, \sigma^2, \lambda) = -\frac{n}{2}\log(\sigma^2) - \sum_{j=1}^{n} \frac{(g_\lambda(y_j) - \underline{x}_j\underline{\beta})^2}{2\sigma^2} + (\lambda-1)\sum_{j=1}^{n}\log(y_j)$$

$$= -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(g_\lambda(y_j) - \underline{x}_j\underline{\beta})^2 + (\lambda-1)\sum_{j=1}^{n}\log(y_j) + \text{const.}$$

Set $\hat{\underline{\beta}}_\lambda = (\underline{X}^T\underline{X})^{-1}\underline{X}^T g_\lambda(y)$ and $\hat{\sigma}^2_\lambda = \frac{1}{n}\sum_{j=1}^{n}(g_\lambda(y_j) - \underline{x}_j\hat{\underline{\beta}}_\lambda)^2$

"Profile" log likelihood for λ

$$\ell(\lambda) = -\frac{n}{2}\log\left(\hat{\sigma}^2_\lambda\right) + (\lambda-1)\sum_{j=1}^{n}\log(y_j) + \text{constant}$$

Recall $\underline{Y} = \underline{X}\underline{\beta} + \underline{\varepsilon}$ where $\mathbb{E}\,\underline{\varepsilon} = \underline{0}$ and $\text{cov}(\underline{\varepsilon}) = \sigma^2 \underline{I}$

$\hat{\underline{\beta}} = (\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{Y}$ is the OLS Estimator.

Gauss-Markov Theorem

Suppose we want to estimate $\psi = \underline{t}^T\underline{\beta}$ where $\underline{t} \in \mathbb{R}^p$. we can estimate ψ as $\hat{\psi} = \underline{t}^T\hat{\underline{\beta}}$. This estimator is linear in $\underline{Y}$ as

$$\hat{\psi} = \underline{t}^T\hat{\underline{\beta}} = \underline{t}^T(\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{Y}$$

This is unbiased since

$$\mathbb{E}(\hat{\psi}) = \mathbb{E}(\underline{t}^T\hat{\underline{\beta}}) = \underline{t}^T\mathbb{E}(\hat{\underline{\beta}}) = \underline{t}^T\underline{\beta}.$$

as $\hat{\underline{\beta}}$ is unbiased for $\underline{\beta}$.

Among linear unbiased estimators of $\psi = \underline{t}^T\underline{\beta}$, do any have smaller variance than $\hat{\psi} = \underline{t}^T\hat{\underline{\beta}}$? No.

Proof: Let $\tilde{\psi} = \underline{c}^T\underline{Y}$ be another linear unbiased estimator of ψ. Then

$$\text{var}(\tilde{\psi}) = \underline{c}^T\text{cov}(\underline{Y})\underline{c} = \underline{c}^T(\sigma^2\underline{I})\underline{c} = \sigma^2\underline{c}^T\underline{c}$$

Since $\tilde{\psi}$ is unbiased

$$\underline{t}^T\underline{\beta} = \mathbb{E}(\tilde{\psi}) = \mathbb{E}(\underline{c}^T\underline{Y}) = \underline{c}^T\mathbb{E}(\underline{Y}) = \underline{c}^T\underline{X}\underline{\beta}$$

so $\underline{t}^T = \underline{c}^T\underline{X}$.

However, $\text{var}(\hat{\psi}) = \underline{t}^T\text{cov}(\hat{\underline{\beta}})\underline{t} = \underline{t}^T(\sigma^2(\underline{X}^T\underline{X})^{-1})\underline{t}$

$$= \sigma^2\underline{t}^T(\underline{X}^T\underline{X})^{-1}\underline{t} = \sigma^2\underline{c}^T\underline{X}(\underline{X}^T\underline{X})^{-1}\underline{X}^T\underline{c}$$

$$= \sigma^2\underline{c}^T\underline{H}\underline{c}$$

Comparing $\text{var}(\hat{\psi})$ with $\text{var}(\tilde{\psi})$

$$\text{var}(\tilde{\psi}) - \text{var}(\hat{\psi}) = \sigma^2\underline{c}^T\underline{c} - \sigma^2\underline{c}^T\underline{H}\underline{c} = \sigma^2\underline{c}^T(\underline{I}-\underline{H})\underline{c}$$

We have seen that $(\underline{I}-\underline{H})$ is idempotent so $(I-H)^2 = I-H$ and it is symmetric so $(\underline{I}-\underline{H})^T = (\underline{I}-\underline{H})$.

$$\text{var}(\tilde{\psi}) - \text{var}(\hat{\psi}) = \sigma^2\underline{c}^T(\underline{I}-\underline{H})^T(\underline{I}-\underline{H})\underline{c} = \sigma^2\|(\underline{I}-\underline{H})\underline{c}\|^2 \geq 0$$